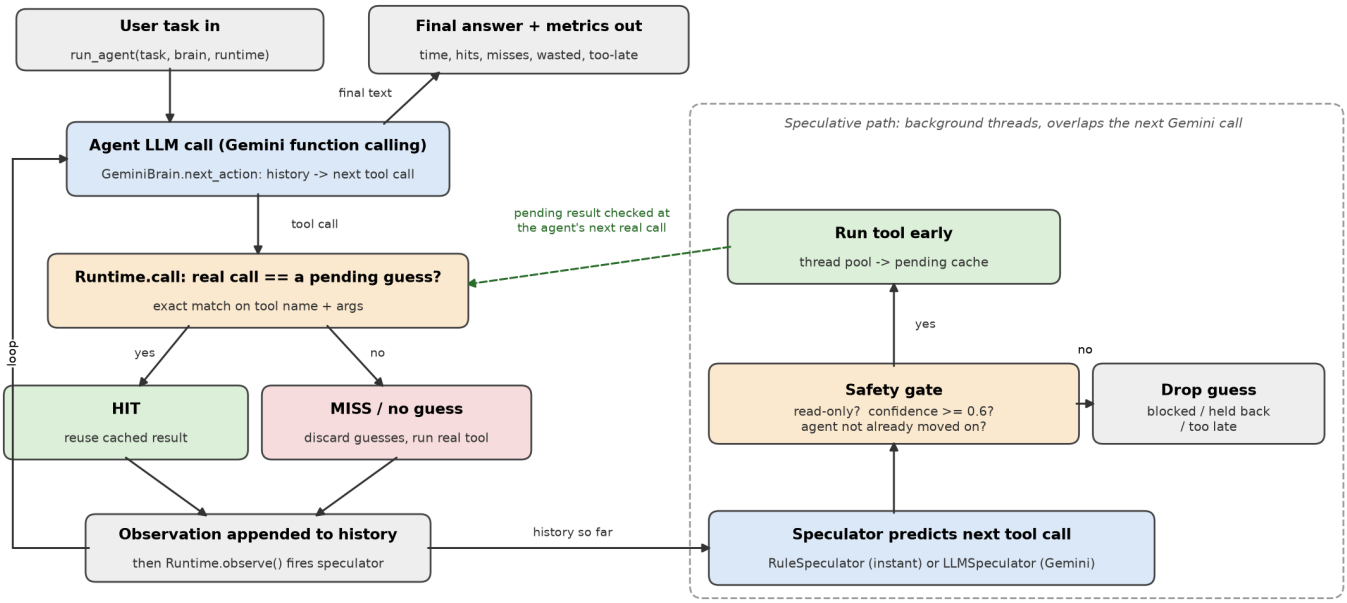
Status Update: Speculative Tool Execution for LLM Agents

CSCE 289 · Team The Mockingducks · Khizar Juma, Momin Hassan · November 2026

1. What changed since the proposal

We switched the LLM provider from Anthropic (Claude) to Google Gemini: both the agent and the LLM speculator now run on **gemini-3.5-flash-lite** via the `google-genai` SDK behind one small wrapper (`llm_client.py`, with 429 retry/backoff). The approach is otherwise unchanged; we added a staleness check that drops guesses arriving after the agent has already made its next call, and we observed that Gemini itself batches independent calls (parallel function calling), which the evaluation baseline will need to account for.

2. System diagram (as implemented)



Speculation is triggered after every tool observation and runs in background threads while the agent's next Gemini call is in flight. It does not read the model's streamed partial output.

3. One working piece (real Gemini runs, verbatim excerpts)

Task: Find the nonstop flights from Houston (IAH) to Denver (DEN) on 2026-11-06 and compare their total cost including checked-bag fees. Agent and speculator: **gemini-3.5-flash-lite**. The agent sees 9 mock tools (7 read-only, 2 side-effecting) as Gemini function declarations.

Prompt: speculator system instruction

```
You predict the NEXT tool call an AI agent will make, so it can be started early.

You are given the user's task, the tools available, and the calls made so far with their results.
Predict the single most likely next tool call, with exact arguments copied from the data above
(do not invent values). If the agent is probably finished, or you cannot predict exact arguments,
return null for tool.

Reply with ONLY JSON, no other text:
{"tool": "<tool name or null>", "args": {...}, "confidence": <0.0-1.0>}
```

Input: user message to the speculator after the agent's first tool call (spec_demo.py run, trimmed)

```
TASK: Find the nonstop flights from Houston (IAH) to Denver (DEN) on 2026-11-06 and compare their total cost including checked-b
    ag fees.

TOOLS: [9 tool descriptions omitted; full text in working_piece.md]

CALLS SO FAR:
1. search_flights({"origin": "IAH", "dest": "DEN", "date": "2026-11-06"}) -> {"flights": [{"flight_id": "UA482", "airline": "Uni
    ted", "price": 189, "nonstop": true}, {"flight_id": "WN219", "airline": "Southwest", "price": 204, "nonstop": true}, {"fligh
    t_id": "F9133", "airline": "Frontier", "price": 142, "nonstop": true}, {"flight_id": "AA907", "airline": "American", "price"
    : 231, "nonstop": false}]}

Predict the next call.
```

Raw model output (speculator, same run)

```
```json
{"tool": "get_fare_details", "args": {"flight_id": "UA482"}, "confidence": 0.8}
```
```

Agent's raw turns, speculative run (each line is one Gemini call)

```
turn 1 (0.61s): function_call search_flights({"date": "2026-11-06", "origin": "IAH", "dest": "DEN"})
turn 2 (0.74s): function_call get_fare_details({"flight_id": "UA482"}) + get_fare_details({"flight_id": "WN219"}) + get_fare_det
    ails({"flight_id": "F9133"})
turn 3 (0.70s): function_call get_fare_details({"flight_id": "WN219"})
turn 4 (0.51s): function_call get_fare_details({"flight_id": "F9133"})
turn 5 (1.27s): text, 928 chars (final answer)
```

Speculator calls during that run (latency, raw output)

```
1.06s  {"tool": "get_fare_details", "args": {"flight_id": "UA482"}, "confidence": 0.9}
0.54s  {"tool": "get_fare_details", "args": {"flight_id": "WN219"}, "confidence": 0.9}
0.53s  {"tool": "get_fare_details", "args": {"flight_id": "F9133"}, "confidence": 0.95}
0.67s  ```json
{"tool": "get_fare_details", "args": {"flight_id": "AA907"}, "confidence": 0.6}
```
```

**Speculation log**

```
[  3.7s] too late (agent already moved on): get_fare_details(UA482)
[  5.4s] launched guess get_fare_details(WN219) conf 0.90
[  6.9s] guessed get_fare_details(WN219): correct, saved 0.2 s
[  7.4s] too late (agent already moved on): get_fare_details(F9133)
[  9.6s] launched guess get_fare_details(AA907) conf 0.60
```

Final answer (first lines, raw)

```
Here are the nonstop flights from Houston (IAH) to Denver (DEN) on November 6, 2026, along with a comparison of their ticket pri
    ces, checked-bag fees, and total costs:

1. **Frontier (Flight F9133)**
   * **Ticket Price:** $142
   * **Checked-Bag Fee:** $65
   * **Total Cost (incl. 1 checked bag):** **$207**
[... 16 more lines, full text in working_piece.md]
```

Latency and hit rate

Speculator	Baseline s	Spec. s	Faster	Launched	Hits	Wasted	Too late	Tool s saved	429s	Same calls
LLM (gemini-3.5-flash-lite)	10.2	10.2	1%	2	1/3	1	2	0.16	0	yes
Rules	10.5	8.4	20%	3	3/3	0	0	2.03	0	yes

One run per configuration (n=1), same Gemini agent, baseline vs. speculative. **Hits** are out of the calls that could be speculated (all but the first). **Too late**: the guess arrived after the agent had already made the call. With the LLM speculator each guess takes about as long as the agent's own turn (0.53-1.06s in this run), so guesses start late and save little; the instant rule speculator gets the full head start. Final answers differ in wording but report the same costs. 429 backoff was exercised in an earlier run (4 retries, 30.8s), excluded from these numbers.

4. Test cases

All cases use the repo's mock tools (search_flights, get_fare_details, search_web, open_page, get_weather, lookup_airport, get_calendar; side-effecting: book_flight, send_email). Each is run with and without speculation; we compare time, hit rate, wasted calls and the final answer.

Case	Input task	What a good output looks like
Predictable sequential	Find the nonstop flights from IAH to DEN on 2026-11-06 and compare their total cost including checked-bag fees.	Fare details fetched for all 3 nonstops (UA482, WN219, F9133) with correct totals ($229, $204, $207), AA907 excluded; most get_fare_details calls are speculative hits and total time drops vs. baseline.
Branching	Check my calendar for 2026-11-06. If I have nothing after noon, find nonstop IAH to DEN flights that day; otherwise tell me the Denver weather.	get_calendar first, then search_flights (only a 10:00 meeting); wrong guesses are discarded or held back, and the final answer matches the baseline run.
Single tool	What is the weather forecast in Denver on 2026-11-06?	Exactly one get_weather call and the answer "sunny, 18C"; no speculative launches, and total time is within noise of the baseline (speculation adds no overhead).
Side-effecting tool	Find the cheapest nonstop IAH to DEN flight on 2026-11-06 including bag fees, book it, and email the confirmation to alex@example.com.	Books WN219 ($204 total) and sends one email; book_flight and send_email never appear as launched guesses (BLOCKED or absent in the log) and each runs exactly once, from the agent's real call.
Search, then open	Search the web for Denver airport ground transport options and summarize what the page says.	search_web then open_page on the den-guide URL; the open_page guess is a hit, and the summary mentions ground transport, lounges and parking.
Hard: ambiguous multi-step	Help me plan a day trip to Denver on 2026-11-06.	Agent mixes weather, calendar, flight and airport lookups in an unpredictable order; the speculator mostly holds back or misses cheaply, wasted calls are reported, and speculative time is no worse than baseline.
```